

Paul Abrams

Toronto, Ontario

Senior Backend Engineer · Node.js / TypeScript · .NET Core / C# · Java · Python · Distributed Systems & Concurrency · REST APIs & API Gateways · SQL & NoSQL · Docker / Kubernetes / Serverless · 15+ Years

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Professional Summary

Senior backend engineer with 15+ years designing, building, and operating highly available, distributed web services in production — from a **load-balanced, geographically distributed** legal case-management platform serving **5,000+ concurrent users nationally, 24/7**, to **cloud-native microservices** on Microsoft Azure. **Language-agnostic by practice:** production **Node.js (Express, NestJS, WebSocket)** and **.NET Core / C#**, with **Java** (an automation framework) and **Python** (LLM-evaluation grading and tooling) rounding out the toolkit alongside **SQL** and **TypeScript**. **Strong REST API design:** designs and consumes **REST APIs between services**, publishes and governs them through an **API gateway (Azure API Management)** with authentication, throttling, and transformation, and generates **living API documentation from JSON-Schema contracts**. Deep **concurrency and fault-tolerance** experience — **queue-based asynchronous fan-out across microservices, transactional concurrency control in PostgreSQL** (`SELECT ... FOR UPDATE`), load balancing across distributed tiers, and a self-built **availability-monitoring suite**. **Security best practices by default:** **OAuth 2.0 / OpenID Connect / PKCE, JWT**, Managed Identities, Key Vault secret hygiene, **TLS / certificate management**, **SonarQube + OWASP ZAP in pipelines**, and **CWE / CVSS-graded security audits**. **Advanced data modeling in SQL (SQL Server, Azure SQL, PostgreSQL) and NoSQL (MongoDB)**. Hands-on with **Docker, Kubernetes / AKS, Red Hat OpenShift, serverless (Azure Functions)**, and **CI/CD** (Azure DevOps, GitHub Actions, Jenkins). Active open-source contributor — **60+ public PRs** including Microsoft's `vsts-team-calendar`, `tfs-cli`, and the **Microsoft Authentication Library .NET** docs.

Professional Experience

Abrams Software Toronto, Ontario (Remote)

August 2025 – Present

Backend / Full-Stack Engineer · Independent

- **TWNR — architected, shipped, and operates a public-facing multiplayer game as a full-stack web application, solo** — owning the entire cycle from development and testing to running the code on production. Dockerized **TypeScript** monorepo: **Node.js (Express) + WebSocket** backend backed by **PostgreSQL**, per-player **JWT auth**, and a **JSON-Schema-generated REST API contract** that doubles as live documentation. **Concurrency-correct multiplayer state** via PostgreSQL transactions (`SELECT ... FOR UPDATE`). Live at twnr.paulabrams.ca.
- **Chess Attack Visualization — React 19 + TypeScript** app consuming the **Lichess.org REST API**, with a **NestJS** auth proxy implementing the **OAuth2 PKCE** flow. Diagnosed a missing feature in `chess.js`, opened [upstream PR #572](#), and published a fork on npm. **Portfolio site** operated as **Astro on GitHub Pages with GitHub Actions CI/CD**.
- **LLM code evaluation** for DataAnnotation.tech and Stellar.ai — prompt fixtures and grading rubrics across **Node / NestJS, .NET / C#, Java, Python, TypeScript, React, Go, Rust, SQL, and MongoDB**, including **security audits with CWE / CVSS-graded inventories**. **60+ public PRs** to outside repos including the **Microsoft Authentication Library .NET** docs and canada-ca/platform-security.

Employment and Social Development, Canada Gatineau, QC

December 2021 – July 2025

Senior Backend Engineer — Cloud-Native Microservices (Benefits Delivery Modernization)

- Enhanced, maintained, and **supported in production a cloud-native microservices** enterprise onboarding application — **.NET Core / C# back end, Blazor front end, Azure SQL, Azure Functions, GraphQL** — serving procurement, HR, and IT-ops personnel. **Designed and consumed**

REST APIs between services and published and governed them through Azure API

Management (API gateway) with authentication, throttling, and transformation policies; integrated enterprise IAM via **Azure AD / Entra (OAuth 2.0, OpenID Connect)**.

- **Concurrency and event-driven fan-out at scale:** Blazor submissions kicked off **serverless Azure Functions** through **queue- and storage-based handoffs — bulk submissions fanning out asynchronously across the microservices** — creating/updating Azure DevOps work items, running pipelines, and provisioning users in on-prem Active Directory (via AAD Connect) by personnel profile.
- **Containerized delivery, orchestration, and CI/CD:** operated end-to-end **Azure DevOps pipelines** for build, **Docker** containerization, **Kubernetes / AKS** deployment, and **Azure Data Factory** (ETL) orchestration. Used **Terraform** (infrastructure-as-code) to provision cloud infrastructure, including **IP-address allocation**. Completed formal **Red Hat OpenShift** training and **resolved OpenShift production-support tickets**; troubleshooted an **Ansible**-driven provisioning pipeline; evaluated **Aqua** container security. For the **AWS-hosted EI application**, deployed a **self-hosted pipeline agent into AWS** (AWS foundational training).
- **Security and observability in production:** migrated the application's downstream identity from rotating secrets (Key Vault + ADO PATs on 90-day rotation) to **Azure Managed Identities** — eliminating the secret-rotation incident class. Implemented **ITSG-33 (NIST-based) controls** against a **PBMM** profile, with **TLS / certificate management** and **Key Vault** secret hygiene. Instrumented and monitored the workloads with **Application Insights, Azure Monitor, and Log Analytics**.
- Designed and shipped **custom web extensions in React, TypeScript, and Node.js** — a build/deploy analytics widget rendering success-rate charts via the ADO Builds REST API, and a people-by-project reporter that **replaced an unusably slow query with an in-memory hashmap**. Ran weekly **demos and design walkthroughs**, mentoring junior developers.

Employment and Social Development, Canada Gatineau, QC August 2019 – November 2021

Development Lead, Testing Services — CI/CD Automation & DevSecOps

- Co-architected and developed an **internal C# / .NET web application** — web UI for authoring structured automation scenarios, **SQL Server** schema design, and a **co-designed REST API** providing bidirectional integration with **Azure DevOps pipelines** (versioned script-fetch endpoints and structured results-push endpoints). Worked alongside a **Jenkins**-based automated-testing helper app under development by the team.
- **Developed automated execution frameworks driven through CI/CD pipelines:** engineered Azure DevOps pipelines that provisioned Windows build agents to run **parallelized automated runs** against deployed applications and fed structured results back into work items. Performed **code review and standards enforcement** for the team.
- **DevSecOps / security scanning:** integrated **SonarQube** static analysis (SAST) into Azure DevOps pipelines and wrote a script that **queried the SonarQube REST API to flag vulnerabilities by keyword and surface them into Azure DevOps**. Ran **OWASP ZAP** dynamic scans (DAST) in pipelines, and evaluated **MicroFocus Fortify / ScanCentral** (SAST) for adoption.
- Drove a tooling consolidation (**LoadRunner → JMeter**) that **cut software licensing costs** while preserving capability, and contributed to **performance-optimization analysis of high-traffic deployed applications**.

Department of Immigration, Refugees, and Citizenship, Canada

Ottawa, ON

August 2017 – August 2019

Team Lead / Technical Lead — Platform & Tooling

- **Technical leadership of a cross-functional team** — led **Agile / Scrum** sprint planning, review, retrospective, daily stand-up, and backlog refinement; introduced poker-planning estimation and capacity forecasting; set engineering standards and drove delivery cadence.
- Implemented and deployed **custom web extensions in React / TypeScript / Node.js** — leading the work hands-on while guiding the team.

- **Mentored developers in SQL Server query optimization** — query tuning, indexing strategy, partitioned tables, and execution-plan analysis against large databases and warehouses; coached on C# / .NET and SSRS.
- Administered departmental **Azure DevOps Server / TFS** instances and heavily customized process templates; managed user/group security and source-control access across multiple project collections.

Department of Justice, Canada Ottawa, ON

December 2015 – August 2017

Acting Lead Developer — iCase / Microsoft Dynamics CRM Modernization

- **Lead developer** on the live **Legal Case Management System (iCase)** — a large **n-tier ASP.NET / C# / ADO.NET** enterprise web application used by **5,000+ lawyers** — while concurrently building its **Microsoft Dynamics CRM** successor. Modified schema and **front-end UI to surface data-migration status in real time**, and adjusted integration stored procedures consumed by the department's SAP-based financial system.
- Engineered **Dynamics CRM plug-ins, custom actions, and workflows in C#** (rewriting legacy stored procedures as C# / LINQ); used **SSIS** for ETL between iCase and Dynamics; produced **entity-relationship diagrams** and **multi-month effort-estimation breakdowns** and applied design patterns (Repository, Strategy, Mediator).
- **Established code-review processes gated on work items** — enforcing maintainability, security, and performance standards — and **mentored junior developers**.

Department of Citizenship and Immigration, Canada Ottawa, ON

April 2014 – December 2015

Senior Software Engineer · Release Engineering Lead

- Hired alongside one other engineer to take over a release-management environment from a contractor team. **Decommissioned over-engineered legacy C# desktop apps** and replaced them with lightweight **PowerShell calling REST APIs** — most contractor tooling was simply deleted and **everything ended up in source control**.
- Implemented and deployed **web extensions in React / TypeScript / Node.js**; maintained a fork of Microsoft's `vsts-team-calendar` extension and contributed back upstream. **Installed, configured, customized, and upgraded** the Azure DevOps Server / TFS platform across dev / UAT / production — complex multi-step upgrades, **Kerberos / SPN configuration** across application tiers, and database-tier migration.

Department of Justice, Canada Ottawa, ON

October 2008 – April 2014

Full-Stack Developer — Legal Case Management System (iCase)

Core developer on the department's flagship **Legal Case Management System (iCase)** — a **distributed, multi-tier ASP.NET / C# enterprise web platform** used by **5,000+ government lawyers and contracted agent firms across Canada**. **High-availability, fault-tolerant, and load-balanced**: deployed across **5 load-balanced web servers** and **11 regionally distributed document servers** nationwide, with role- and object-based security and integration to records-management, financial, and email systems, running **24/7**.

- **Optimized a high-traffic backend**: diagnosed a homepage report taking 10 s uncached — profiled with SQL Profiler, Report Execution logs, and execution plans, then replaced live joins with a nightly-populated flat table for a **50x speedup uncached (3-4x cached)**. Built a **self-initiated availability-monitoring suite** across the distributed web and document servers, and diagnosed and resolved a **COM Interop memory leak** affecting the document tier. Co-led the **SQL Server 2005 → 2012** and **.NET Framework 2.0 → 4.0** upgrades.
- **Java automation-framework development**: architected a **Java-based framework (IBM Rational Functional Tester)** that **parsed English-language scenario documents into executable scripts with round-trip code generation**, plus a companion manager in **C# / SQL Server**. Replaced a labour-intensive **two-week, 20-25-person manual pre-release validation cycle** with automated execution reviewed by one person in five days — **98% reduction in effort, \$160K saved per release cycle**.

Education

Certifications & Awards

Professional Scrum Master I (Scrum.org) · ITIL v3 Foundation (Loyalist Certification Services) · DOJ Team Merit Award 2012; Team Merit Award 2011; Team Spirit Award 2009

Technical Skills

Languages (language-agnostic toolkit): TypeScript / JavaScript (Node.js), C#, Java, Python, SQL (T-SQL), PowerShell, Go (light), XML / XSLT

Backend Frameworks: Node.js, Express, NestJS · .NET Core / .NET 6 / 8, ASP.NET Core, ADO.NET, Entity Framework, LINQ · Spring Boot (familiarity)

Distributed Systems & Concurrency: cloud-native microservices, event-driven & asynchronous messaging (queue- / storage-triggered fan-out), transactional concurrency control (PostgreSQL SELECT ... FOR UPDATE, SQL Server isolation levels), load balancing across distributed tiers, 24/7 high availability & fault tolerance, availability monitoring

APIs: RESTful API design between services · API gateway (Azure API Management) — authentication, throttling, transformation · API documentation from JSON-Schema contracts (self-documenting) · API security (OAuth 2.0 / OpenID Connect / PKCE, JWT) · GraphQL, WebSocket

Network Protocols: HTTP / HTTPS, TLS / certificate management, WebSocket, TCP/IP fundamentals, DNS & IP provisioning (Terraform), Kerberos / SPN configuration across multi-tier deployments

Databases & Data Modeling: SQL — MS SQL Server 2005 → 2019 (schema design, stored procedures, indexing, partitioning, execution-plan tuning), Azure SQL, PostgreSQL · NoSQL — MongoDB (working familiarity)

Cloud, Containers & Serverless: Docker, Kubernetes / AKS, Red Hat OpenShift · Microsoft Azure — App Service, Azure Functions (serverless), AKS, Azure SQL, API Management, Key Vault, Managed Identities, Entra / AAD, Storage · AWS — self-hosted pipeline-agent deployment + foundational training · GCP — TWNR production hosting

DevOps / CI-CD: Azure DevOps Pipelines (YAML), GitHub Actions, Jenkins · Terraform (IaC), Ansible · build, automated test, containerized deploy · Azure Data Factory (ETL)

Security / DevSecOps: SonarQube (SAST — pipeline integration + REST-API-driven vulnerability queries), OWASP ZAP (DAST) · OAuth 2.0 / OpenID Connect / PKCE, JWT, Entra / AAD IAM, Managed Identities · Key Vault secret hygiene · ITSG-33 / PBMM controls, CWE / CVSS-graded security audits

Observability: Azure Monitor, Application Insights, Log Analytics · Prometheus / Grafana (familiarity) · self-built distributed availability-monitoring suite

Process / SDLC: Agile / Scrum (PSM I) — daily stand-ups, sprint planning, review, backlog refinement · full ownership from development/testing through code review to production support · mentoring, technical documentation · ITIL v3 Foundation